WEEK – 3

b) Develop a Java program to create an interface ParkingCharge containingthe methods calculateCharge() and display ParkingDetails(). Create a class VehicleParking with members Vehicle Number, Vehicle Type, and ParkingHours. Derive the classes TwoWheeler and FourWheeler from VehicleParking and implement the interface. Calculate the parking charge as follows: For Two Wheeler, charge Rs. 20 per hour for the first two hours and Rs. 10 per hour for each additional hour. For FourWheeler, charge Rs. 50 per hour for the first two hours and Rs. 30 per hour for each additional hour.Display vehicle details and total parking charge. Demonstrate runtime polymorphism using a ParkingCharge reference.

PROGRAM

```
interface ParkingCharge {
    double calculateCharge();
    void displayParkingDetails();
}


class VehicleParking {
    String vehicleNumber;
    String vehicleType;
    int parkingHours;


    VehicleParking(String number, String type, int hours) {
        vehicleNumber = number;
        vehicleType = type;
        parkingHours = hours;
    }
}
```

```
class TwoWheeler extends VehicleParking implements ParkingCharge {

    TwoWheeler(String number, int hours) {
        super(number, "Two Wheeler", hours);
    }

    public double calculateCharge() {
        if (parkingHours <= 2)
            return parkingHours * 20;
        else
            return 2 * 20 + (parkingHours - 2) * 10;
    }

    public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours);
        System.out.println("Parking Charge: Rs. " + calculateCharge());
        System.out.println();
    }
}

class FourWheeler extends VehicleParking implements ParkingCharge {
```

```
    FourWheeler(String number, int hours) {
        super(number, "Four Wheeler", hours);
    }

    public double calculateCharge() {
        if (parkingHours <= 2)
            return parkingHours * 50;
        else
            return 2 * 50 + (parkingHours - 2) * 30;
    }

    public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours);
        System.out.println("Parking Charge: Rs. " + calculateCharge());
        System.out.println();
    }
}

public class Main {
    public static void main(String[] args) {

        ParkingCharge p;
```

```
        p = new TwoWheeler("KA01AB1234", 4);
        p.displayParkingDetails();

        p = new FourWheeler("KA02CD5678", 3);
        p.displayParkingDetails();

        System.out.println("NAME: Deekshitha H S");
        System.out.println("USN: 1BM26CS605-T");
    }
}
```

OUTPUT

```
Vehicle Number: KA01AB1234
Vehicle Type: Two Wheeler
Parking Hours: 4
Parking Charge: Rs. 60.0

Vehicle Number: KA02CD5678
Vehicle Type: Four Wheeler
Parking Hours: 3
Parking Charge: Rs. 130.0

NAME: Deekshitha H S
USN: 1BM26CS605-T
```